

PALAK TANWAR

tanwar.pa@northeastern.edu | 617-404-6023 | linkedin.com/in/ptanwar | github.com/palaktanwar | ptanwar.me

Computer Engineer and Data Analytics Engineering MS candidate (GPA 3.94) who builds software end to end back-end services, REST APIs, and data pipelines with a CS foundation in OOP, data structures, and algorithms, and a specialization in Machine Learning/GenAI systems. Seeking full-time opportunities from January 2027.

EDUCATION

Northeastern University - MS, Data Analytics Engineering (GPA 3.94/4.0) Boston, MA | Expected Dec 2026

Coursework: Natural Language Processing, Machine Learning Operations, Data Management, Advanced Algorithms

Thapar Institute of Engineering & Technology - BE, Computer Engineering India | Jun 2019

Coursework: Artificial Intelligence, Parallel & Distributed Computing, Image Processing

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (MySQL, PostgreSQL), R, NoSQL (MongoDB, Neo4j)

CS Foundations: Object-oriented design, data structures & algorithms, REST, client server architecture, data modeling & validation

Backend & Cloud: FastAPI, MySQL, PostgreSQL, MongoDB, Neo4j, Docker, Git, Unix/Linux, GCP (BigQuery, GCS), Airflow, MLflow

ML/GenAI & Viz: LLMs (Google Gemini), NLP, prompt engineering, NL to SQL, PyTorch, TensorFlow, Scikit-learn, XGBoost, GloVe/embeddings, SHAP, SMOTE, PowerBI, Quicksight, Tableau, Grafana, dashboards

PROJECTS

Sentiment Classification with Classical & Neural NLP Models

Jun 2026

- Implemented perceptron and logistic-regression text classifiers **from scratch** (no ML libraries) and improved **test accuracy from 76% to 85%** via bigram and custom feature engineering and learning rate tuning.
- Built neural models in **PyTorch**, a Deep Averaging Network with pretrained **GloVe** embeddings and a **CNN** with multi-width convolution filters, reaching **90% test accuracy**, diagnosing overfitting from loss curves.
- Re-engineered training to mini batch on GPU for a **20x speedup** with no accuracy loss performance optimization and low-level debugging.

Corporate Financial Distress Early Warning System – Foresight ML | 1st Place, MLOps Expo @ Google Cambridge

Apr 2026

- Awarded **1st place out of 30 teams** at the **6th MLOps Expo hosted at Google Cambridge**, judged on system clarity, production-readiness, deployment pipelines, and engineering depth; owned the end-to-end deployment layer: model versioning, inference schema, and manifest-based lineage tracking for fully traceable batch inference.
- Built an end-to-end pipeline that flags companies at risk of financial distress **6-12 months ahead**, fusing SEC filings (10-K/10-Q) with FRED macro indicators for **10,000+ U.S. public companies**.
- Engineered **45+ features** in **BigQuery** and pandas (liquidity, leverage, macro-interaction terms) over a large-scale data processing pipeline; trained **XGBoost** and **LSTM/GRU** models tuned for high recall so few at risk firms slip through.
- Applied **SHAP** for per-prediction explainability and ran bias checks across sectors and economic regimes, keeping the risk score auditable, not a black box.

LLM-Powered MLOps Platform & Natural Language Query Assistant

Dec 2025

- Built the backend for a self-serve query service: a **FastAPI** layer wrapping a **Google Gemini** natural language to SQL assistant in a **multi-stage validation** pipeline that enforces read only access and blocks unsafe queries client server API design plus input validation.
- Designed a hybrid **MySQL + Neo4j** backend tracking datasets, experiments, Git commit hashes, and model artifacts, giving every result reproducible lineage.
- Shipped a **Streamlit** dashboard that uses LLMs to auto generate plain English summaries of model performance and KPIs for cross functional teams.

Patient Cohort & Readmission Analytics - MIMIC-III

Feb 2025

- Worked with real ICU patient records under strict privacy handling to segment patients and **predict hospital readmissions**, an operational metric that drives healthcare cost and care decisions.
- Reduced high dimensional clinical data with **PCA** and **t-SNE**, used **K-means** and hierarchical clustering to surface distinct patient cohorts.

WORK EXPERIENCE

Chaitanya Charitable Trust - Data Analyst & Strategist

Gujarat, India | Jun 2020 - Jul 2024

- Owned data end to end for community programs: pulled and joined data with **SQL**, cleaned messy survey inputs in **Python** and spreadsheets, and turned them into trusted **reporting dashboards** leadership relied on.
- Translated vague, on the ground needs from **1,000+ stakeholders** into clear data definitions and metrics, then used the analysis to decide which campaigns to run and where to focus, growing audience reach **35%**.
- Drove cross functional buy in by explaining the reasoning behind the numbers in plain language, lifting program participation **30%**.